Customer Churn Prediction Report

Introduction

This project focused on predicting customer churn in a telecommunications company using machine learning. The goal was to identify customers likely to leave the service so that the company could intervene early with retention strategies. Churn prediction is a valuable business tool because retaining existing customers is usually more cost-effective than acquiring new ones

Data Understanding

The Telco Customer Churn dataset was used for the analysis. It contains customer demographic details, account information, and service usage variables, with `Churn` as the target column indicating whether a customer left within the last month. The dataset also includes variables such as tenure, monthly charges, and total charges, which are commonly associated with churn behavior.

Data Preparation

The dataset was first inspected to check its structure, missing values, and data types. The `TotalCharges` column required conversion to numeric because it may be read as text, and missing values were handled appropriately. The target variable `Churn` was encoded into binary form so that it could be used for classification modeling#

Exploratory Data Analysis

Exploratory analysis showed the distribution of churn across customers and helped reveal patterns in tenure and monthly charges. Visualizations such as churn counts, tenure distributions, and monthly charges by churn category provided insight into how customer behavior may relate to churn. These charts supported the business objective by highlighting customer groups that may need retention attention

Feature Engineering and Encoding

To improve model learning, additional features were created from the existing variables, including average monthly charges and a long-tenure indicator. Categorical variables were converted into numerical format using one-hot encoding, allowing the machine learning model to process them effectively. The target variable was separated from the predictors before model training to avoid data leakage

Model Training and Prediction

A classification model was trained on the prepared dataset to predict whether a customer would churn. The data was split into training and testing sets, and the model was fit on the training data before generating predictions on the test set. This approach allowed the model's performance to be evaluated on unseen data, which is important for judging real-world usefulness.

Model Evaluation

The model was evaluated using accuracy, precision, recall, F1 score, and a confusion matrix. These metrics provide a balanced view of performance, especially in churn prediction where recall is important because missing a likely churn customer can lead to business loss. The confusion matrix also helped show how many churn and non-churn cases were correctly or incorrectly classified.

Conclusion

The analysis demonstrated that machine learning can be used to predict customer churn from billing and service-related information. The model can help the telecommunications company identify at-risk customers and target them with retention campaigns before they leave.